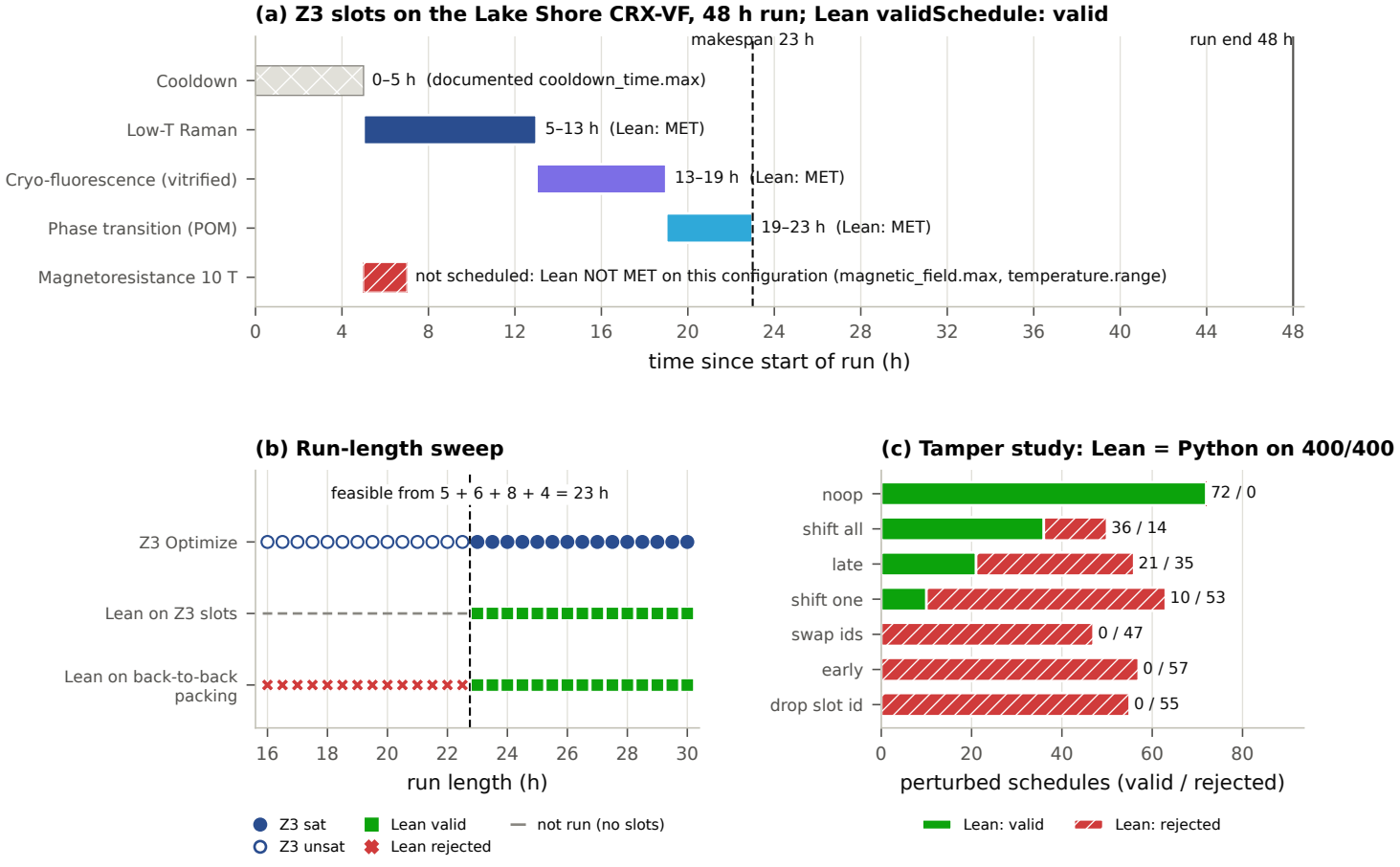
(a) Z3 slots on the Lake Shore CRX-VF, 48 h run; Lean validSchedule: valid
makespan 23 h
run end 48 h
Cooldown
0–5 h (documented cooldown_time.max)
Low-T Raman
5–13 h (Lean: MET)
Cryo-fluorescence (vitrified)
13–19 h (Lean: MET)
Phase transition (POM)
19–23 h (Lean: MET)
Magnetoresistance 10 T
not scheduled: Lean NOT MET on this configuration (magnetic_field.max, temperature.range)
0
4
8
12
16
20
24
28
32
36
40
44
48
time since start of run (h)
(b) Run-length sweep
feasible from 5 + 6 + 8 + 4 = 23 h
Z3 Optimize
Lean on Z3 slots
Lean on back-to-back packing
16
18
20
22
24
26
28
30
run length (h)
Z3 sat
Z3 unsat
Lean valid
Lean rejected
not run (no slots)
(c) Tamper study: Lean = Python on 400/400
noop
72 / 0
shift all
36 / 14
late
21 / 35
shift one
10 / 53
swap ids
0 / 47
early
0 / 57
drop slot id
0 / 55
0
20
40
60
80
perturbed schedules (valid / rejected)
Lean: valid
Lean: rejected